

JACKSON C. TURNER

QUANTUM RESEARCH SCIENTIST | APPLIED MATHEMATICIAN

New York, NY | me@jacktrnr.com | jacktrnr.com | github.com/jacktrnr

Education

Columbia University

MS/Ph.D. in Applied Mathematics (GPA: 3.97)

June 2026

New York, NY

Engineering Presidential Fellow. Dissertation: Resonance-induced nonlinear bound states and their orbital stability. Advisor: M. I. Weinstein.

University of Utah

Honors B.S. in Applied Mathematics & Physics (GPA: 3.95)

May 2021

Salt Lake City, UT

Presidential Scholar. Developed mesh-free PDE eigensolvers

Experience

Harmoniqs

Brooklyn, NY

Quantum Research Scientist

Jun 2026 – Present

- Design and optimize quantum control protocols for neutral-atom and trapped-ion hardware
- Lead technical collaboration meetings with academic partners (Stanford, UW, MIT) and author research grant proposals
- Drive investor presentations and contribute directly to company fundraising

Atlas Fusion

Remote

Mathematical Physicist (Contract)

Feb – Jun 2026

- Built numerical solvers in Julia for plasma stability problems

Columbia University, APAM Department

New York, NY

Graduate Research Assistant

May 2022 – Present

- Connected quantum scattering theory to nonlinear bifurcation theory, establishing how resonances in linear wave operators generate nonlinear bound states; paper under review at *Nonlinearity*
- Built numerical solvers for scattering, bifurcation, and dynamical systems problems with $< 10^{-8}$ tolerances; invited talk, Simons Collaboration on Extreme Wave Phenomena; Best Poster, Joint Alabama–Florida Conference (2025)

Zap Energy, Theory & Modeling Division

Everett, WA

Research Intern

Jun – Aug 2025

- Achieved $720\times$ speedup (4 hours $\rightarrow$ 20 seconds) building eigenvalue solver in Julia for MHD linear stability analysis; used predictor-corrector continuation with arclength parameterization to track embedded eigenvalues through bifurcations
- Performed spectral analysis of axisymmetric ideal MHD equilibria; studied onset of Kelvin–Helmholtz and shear-driven Mack-mode instabilities via bifurcation theory and complex analysis
- Identified novel plasma instability mechanism in sheared-flow Z-pinch configurations; preprint at arXiv:2608.00291

Volunteer Program Leader

Hong Kong

Full-time Service

2015 – 2017

- Led teams of 10–20 volunteers across multiple districts, managing goal-setting and training; fluent in Cantonese

Technical Skills

Mathematics Spectral & scattering theory, nonlinear dispersive PDE, bifurcation analysis & numerical continuation, Lyapunov & orbital stability of Hamiltonian systems, variational calculus, functional analysis, perturbation methods, complex analysis, constrained optimization

Physics Quantum optimal control (neutral atoms, trapped ions, optical tweezers), quantum mechanics, magnetohydrodynamics & plasma stability, Hamiltonian dynamics, dispersive wave propagation

Computation High-accuracy eigenvalue & scattering solvers, spectral methods for PDEs, predictor-corrector continuation & bifurcation tracing, nonlinear optimization, mesh-free PDE methods, GPU-accelerated scientific computing

Programming Julia (primary), Python (NumPy, SciPy), MATLAB; familiar with Fortran, PyTorch

Publications

- Crews, D. W. & J. C. Turner. “[On the ideal stability of the sheared-flow Z pinch.](#)” arXiv:2608.00291, 2026.
- Turner, J. C. & M. I. Weinstein. “Lyapunov stability of resonance-induced nonlinear bound states.” In preparation, 2026.
- Turner, J. C. & M. I. Weinstein. “[Resonance-induced nonlinear bound states.](#)” arXiv:2510.19538, 2025.
- Kao, C.-Y., B. Osting & J. C. Turner. “Flat Tori with Large Laplacian Eigenvalues.” *SIAM J. Appl. Algebra Geom.*, 2023.
- Turner, J. C., E. Cherkaev & D. Wang. “[Expansion Method for Laplace–Beltrami Eigenfunctions.](#)” arXiv:2210.10982, 2022.